FLYBOAT

Guiyang High-Tech Yige Electronic Co., Ltd.

20S Series Metal Oxide Varistors — Datasheet

Features

- Centre melt-through under power-frequency overload: the melt-through (puncture) point is engineered to occur at the centre of the disc, underneath the copper electrode, so heat is absorbed and conducted straight to the disconnection point — the chip resists melt-through longer and a mating thermal disconnector trips faster.
- For top grades the melt-through point is confined within a 0.5 cm radius of the chip centre, converting a random failure location into a controlled one.
- If the chip does melt through, the flame jet is shielded by the copper electrode, which conducts the heat to the trip point and drives the thermal disconnector open.
- Exceptional damp-heat resistance: under a customer's severe boil test (30 min in boiling water, cool, measure, re-boil; 6 cycles required with no significant change in varistor voltage or leakage), Yige product has reached 8 cycles with small-current parameters essentially unchanged.
- Surge and energy headroom (standard range): In up to 7 kA, Imax up to 15 kA, 2 ms energy absorption up to 485 J.
- Long, verifiable service life (Arrhenius law): verified minimum expected life of 110 years at 40 °C, 11 years at 65 °C, 0.7 years at 95 °C upper ambient temperature (115 °C / 1,000 h ageing basis).
- High TOV withstand: at a temporary overvoltage of 1.45 × Uc the chip stays thermally stable for more than 45 min (r.m.s.), with no damage.

Model naming: 20S K 431 — 20 = nominal size (mm), S = square; K = varistor-voltage tolerance ±10 % (J = ±5 %); 431 = varistor-voltage code (first two digits significant, last digit = number of zeros; 431 = 430 V). Further suffixes give the lead/structure form; -U = uncoated bare electrode disc (no mark = epoxy-coated by default).

1. Electrical characteristics — standard range

U1mA = varistor voltage at DC 1 mA; ΔU1mA = tolerance; UAC / UDC = maximum continuous operating voltage; In = nominal discharge current (8/20 µs); Imax = maximum discharge current (8/20 µs); UP = clamping voltage at In (reference value–ratio); ETM = maximum energy absorption (2 ms). Certification marks reproduce the catalog ratings table.

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)	UL	C-UL	TÜV
20SK101	100	±10	60	85	7	15	360-3.6	70	✓	✓	—
20SK121	120	±10	75	100	7	15	400-3.6	85	✓	✓	—
20SK151	150	±10	95	125	7	15	450-3.0	108	✓	✓	—
20SK181	180	±10	115	150	7	15	490-2.8	125	✓	✓	—
20SK201	200	±10	130	170	7	15	520-2.6	140	✓	✓	—
20SK221	220	±10	140	180	7	15	550-2.6	160	✓	✓	—

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)	UL	C-UL	TÜV
20SK241	240	±10	150	200	7	15	600-2.6	170	✓	✓	—
20SK271	270	±10	175	225	7	15	680-2.6	200	✓	✓	—
20SK301	300	±10	195	250	7	15	770-2.6	220	✓	✓	—
20SK331	330	±10	210	270	7	15	830-2.55	240	✓	✓	—
20SK361	360	±10	230	300	7	15	910-2.55	270	✓	✓	—
20SK391	390	±10	250	320	7	15	1000-2.55	300	✓	✓	—
20SK431	430	±10	275	350	7	15	1050-2.45	340	✓	✓	—
20SK471	470	±10	300	385	7	15	1100-2.4	360	✓	✓	—
20SK511	510	±10	320	410	7	15	1150-2.35	390	✓	✓	—
20SK561	560	±10	350	450	7	15	1250-2.35	420	✓	✓	—
20SK621	620	±10	385	505	7	15	1450-2.3	460	✓	✓	—
20SK681	680	±10	420	560	7	15	1600-2.3	460	✓	✓	—
20SK711	710	±10	440	585	7	15	1650-2.3	470	✓	✓	—
20SK751	750	±10	460	615	7	15	1700-2.3	475	✓	✓	—
20SK781	780	±10	485	640	7	15	1750-2.3	480	✓	✓	—
20SK821	820	±10	510	670	7	15	1800-2.3	485	✓	✓	—

2. Electrical characteristics — low-residual-voltage (low-clamping) range

Same column definitions as Table 1; optimised for lower clamping voltage at In.

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)	UL	C-UL	TÜV
20SK431	430	±10	275	350	7	15	990-2.2	340	✓	✓	—
20SK471	470	±10	300	385	7	15	1070-2.2	360	✓	✓	—
20SK511	510	±10	320	410	7	15	1150-2.2	390	✓	✓	—
20SK561	560	±10	350	450	7	15	1250-2.15	420	✓	✓	—
20SK621	620	±10	385	505	7	15	1350-2.1	460	✓	✓	—
20SK681	680	±10	420	560	7	15	1520-2.1	460	✓	✓	—
20SK711	710	±10	440	585	7	15	1600-2.1	470	✓	✓	—
20SK751	750	±10	460	615	7	15	1700-2.1	475	✓	✓	—
22SK781	780	±10	485	640	7	15	1760-2.1	480	✓	✓	—
22SK821	820	±10	510	670	7	15	1800-2.08	485	✓	✓	—

Notes: the last two rows of this table are misprinted in the catalog with a 22SK prefix (“22SK781” / “22SK821”); by position and values they are the 20S rows — shown as printed.

3. Dimensions and thickness

Bare electrode disc thickness (mm), standard & low-clamping ranges — 101: 2.7 · 121: 2.7 · 151: 2.7 · 181: 2.8 · 201: 2.9 · 221: 3.0 · 241: 3.1 · 271: 3.3 · 301: 3.5 · 331: 3.6 · 361: 3.8 · 391: 4.0 · 431: 4.1 · 471: 4.4 · 511: 4.6 · 561: 4.9 · 621: 5.2 · 681: 5.2 · 711: 5.7 · 751: 6.0 · 781: 6.3 · 821: 6.5.

Encapsulated thickness Tmax (mm), standard & low-clamping ranges — 101: 3.8 · 121: 3.8 · 151: 3.8 · 181: 3.9 · 201: 4.0 · 221: 4.1 · 241: 4.2 · 271: 4.4 · 301: 4.6 · 331: 4.7 · 361: 4.9 · 391: 5.1 · 431: 5.2 · 471: 5.5 · 511: 5.7 · 561: 6.0 · 621: 6.3 · 681: 6.5 · 711: 6.6 · 751: 7.1 · 781: 7.3 · 821: 7.5.

Dimension drawings for each package form are available on request.